1 A student investigates the extension of a spring supporting a wooden strip, as shown in Fig. 1.1.

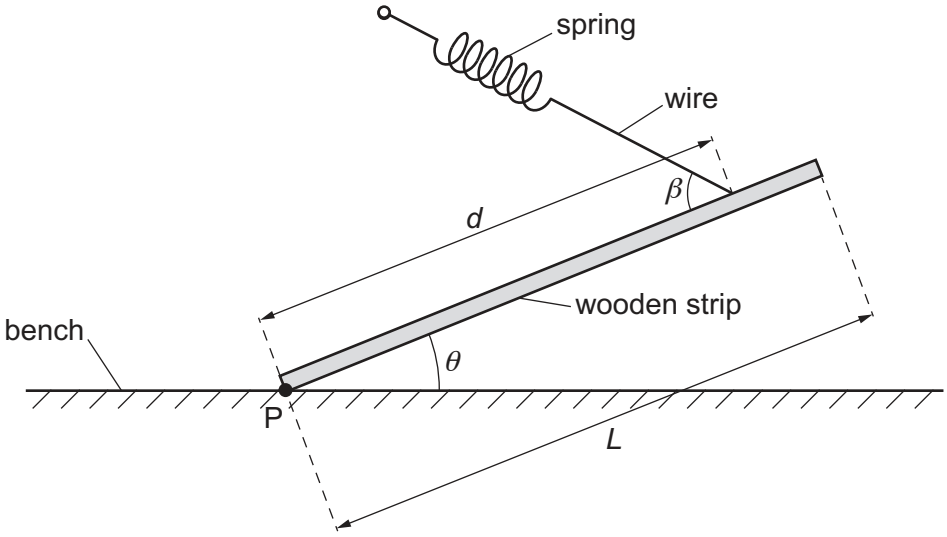

Fig. 1.1

One end of the strip is at point P. The strip has length L and is at an angle θ to the bench.

The spring is attached to the strip by a wire at a distance d from point P. The wire is at an angle β to the strip. The spring has extension x.

It is suggested that the relationship between x and θ is

$$\frac{WL}{2}\cos\theta = kxd\sin\beta$$

where k is the spring constant of the spring and W is a constant.

Design a laboratory experiment to test the relationship between x and θ.
Explain how your results could be used to determine a value for W.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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